

Web Development Foundations

Sample / Practice Question Paper – 1

Mid Term Examination – Practice Paper

Odd Semester 2026–27

Program (Branch), Year, Semester:

Subject Code & Subject Name:

Time: 2 Hours

Maximum Marks: 30

Instructions for Students

1. Attempt all questions.
2. Write syntactically correct HTML/CSS wherever required.
3. Use suitable and meaningful class/id names.
4. Assume reasonable values wherever necessary.
5. Use semantic HTML elements wherever applicable.
6. For CSS questions, write only the essential CSS required to achieve the given result.
7. Clearly mention media queries wherever responsive behavior is required.

Section – A

Attempt All Questions 5 × 3 = 15 Marks

Q1. Semantic HTML – 3 Marks

A university **library webpage** needs a semantic structure containing: A header with the library name A navigation menu containing **Books, Journals, Membership and Contact** A main section containing a heading and a short paragraph An image of the library A footer Write the essential HTML structure using appropriate semantic elements and meaningful attributes for the image and navigation links.

CO2 | BL: A | KL: P

Q2. HTML Form – 3 Marks

Create the essential HTML for a **hostel application form** that collects: Student Name Email Mobile Number Hostel Type using radio buttons Room Preference using a select menu Facilities Required using checkboxes Address using textarea Submit button Use appropriate labels, input types and **required** validation wherever suitable.

CO2 | BL: A | KL: P

Q3. CSS Box Model – 3 Marks

A <div> element has: Width = **300px**, Padding = **10px** on each side, Border = **4px** on each side, and Margin = **20px** on each side.

(a) Calculate the total occupied width using the default **content-box** model.

(b) Write the CSS declaration that changes the element to the **border-box** model.

CO3 | BL: A | KL: P

Q4. CSS Grid and Responsive Design – 3 Marks

A **photo section** contains six photos inside a container. Write CSS to: Display the photos in **three equal columns** on screens of 700px and above. Display them in **two columns** below 700px. Maintain a **20px gap** between the photos. Use CSS Grid and a suitable media query.

CO3 | BL: A | KL: P

Q5. Positioning, Transition and Transform – 3 Marks

A **course card** contains a small “**Popular**” label. The label should remain attached to the bottom-left corner of the card. When the user moves the mouse over the card: The card should move slightly upward. The card should become slightly larger. The effect should occur smoothly. Write the essential CSS using **position: relative**, **position: absolute**, **transition**, and a suitable 2D **transform**.

CO3 | BL: A | KL: P

Section – B

Attempt All Questions 3 × 5 = 15 Marks

Q6. Scenario-Based HTML Task – 5 Marks

Design the essential HTML structure for an **Online Book Store webpage**. The webpage must contain: A semantic header with the website name. A navigation menu containing suitable links. A main section containing: A welcome/introductory section A list of book categories An image of a featured book A table showing at least three books with their title, author and price A customer enquiry form containing at least **four different input types**. A footer containing copyright information. Use appropriate semantic elements and meaningful attributes.

CO2 | BL: C | KL: P

Q7. CSS Layout Task – 5 Marks

Consider the following HTML structure for a **College Department Dashboard**:

```
<div class="dashboard">
  <header>Computer Science Department</header>
  <nav>
    <a href="#">Home</a>
    <a href="#">Faculty</a>
    <a href="#">Courses</a>
    <a href="#">Contact</a>
  </nav>
  <main>
    <div class="department-card">Faculty</div>
    <div class="department-card">Courses</div>
    <div class="department-card">Labs</div>
    <div class="department-card">Events</div>
  </main>
  <footer>Department of Computer Science</footer>
</div>
```

Write CSS to:

- (a) Create the overall page layout using **CSS Grid** with separate areas for header, navigation, main and footer.
- (b) Arrange the four department cards using **Flexbox**.
- (c) Add suitable padding, border, margin and gap.
- (d) Apply **box-sizing: border-box** appropriately.
- (e) Make the navigation links appear in a horizontal row using Flexbox.

Briefly explain the major CSS properties used.

CO3 | BL: An | KL: P

Q8. Integrated Responsive Web Design – 5 Marks

A **movie webpage** contains a navigation bar and **six movie cards**. Write the essential CSS to: Display the movie cards in **three columns** on desktop screens. Display the cards in **one column** on screens below **600px** using a media query. Use **Flexbox** to arrange the title, description and button inside each movie card. Add a smooth hover effect that slightly **rotates and scales** the card using **transition** and **transform**. Create a simple continuous **@keyframes** animation for a **"Now Showing"** badge. Finally, state the purpose of the major CSS properties used in your solution.

CO3 | BL: C | KL: P

Practice Checklist

- Semantic HTML: header, nav, main, section, article and footer
- Links and images with meaningful attributes
- Forms, labels and different input types
- required validation, tables and lists
- CSS Box Model and box-sizing
- CSS Grid and grid-template-columns
- Flexbox and gap
- Relative and absolute positioning

- CSS transition and 2D transform
- Media queries and @keyframes animation
- Combining multiple CSS concepts in scenario-based questions